U.S. DEPARTMENT OF AGRICULTURE

Iron Pasture

Milk Safety Compliance Portal

● Sovereign AI — No data leaves this environment

Powered by Tanzu Platform + Tanzu Data Intelligence

THE PROBLEM

Dairy processors operating under USDA Grade A certification generate continuous safety data — pasteurization temperatures, bacterial counts, phosphatase results — that must be manually cross-referenced against the Pasteurized Milk Ordinance and transcribed into federal inspection forms. A single missed threshold or transcription error can trigger a product recall, license suspension, or federal enforcement action.

AI can close this gap — but plant operational data and process parameters cannot leave the facility. Any solution must run entirely on-premises, produce explainable outputs, and generate a defensible audit trail for USDA regulators.

Hours
Typical time to prepare a NCIMS inspection report manually

< 60 sec
Time from sensor submission to drafted compliance report

7
PMO regulatory sections evaluated per batch submission

0 bytes
Plant data transmitted to a public API

THE SOLUTION

Iron Pasture is a sovereign AI compliance portal that accepts raw plant sensor readings and automatically produces a pre-filled NCIMS 2359 inspection report — grounded in the full text of the Pasteurized Milk Ordinance, with every regulatory citation and compliance decision captured in an immutable audit log.

Plant operators submit data. The AI does the regulatory reasoning. Human inspectors make the final call.

Bronze Ingest
Raw sensor readings
+ PMO document chunks
(append-only)

→

Silver Enrichment
Normalized readings
PMO threshold flags
Top-k passage retrieval

→

Gold Draft
NCIMS 2359 draft
Full audit trail
Model + prompt version

→

Inspector Review
Human adjudication
Approve / Return / Escalate

VMWARE & TANZU PRODUCTS

- VMware Cloud Foundation (VCF) Private AI Foundation
- Tanzu Platform (Cloud Foundry)
- Tanzu Data Intelligence (Greenplum + pgvector)
- Tanzu GenAI Tile (sovereign LLM inference — no external API calls)

Iron Pasture demonstrates that regulated industries don't have to choose between AI-powered efficiency and data sovereignty. The compliance intelligence stays inside the perimeter. The paperwork takes care of itself.

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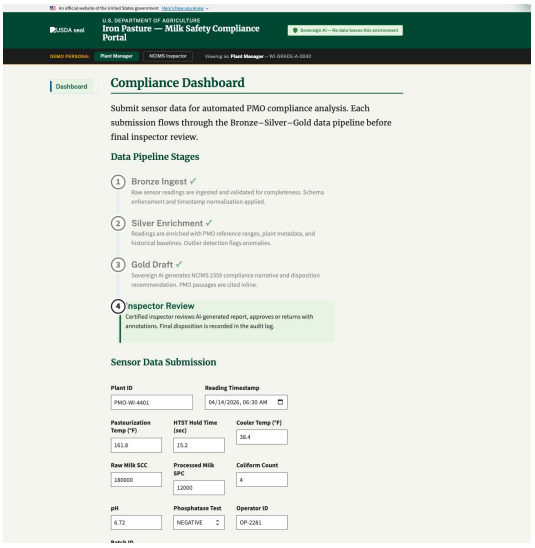
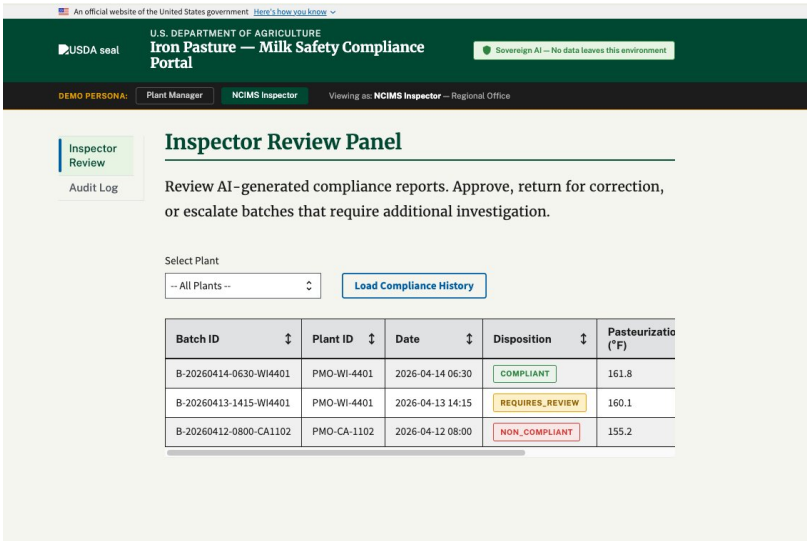
APPLICATION SCREENSHOTS

Fig. 1 — Plant Manager Dashboard



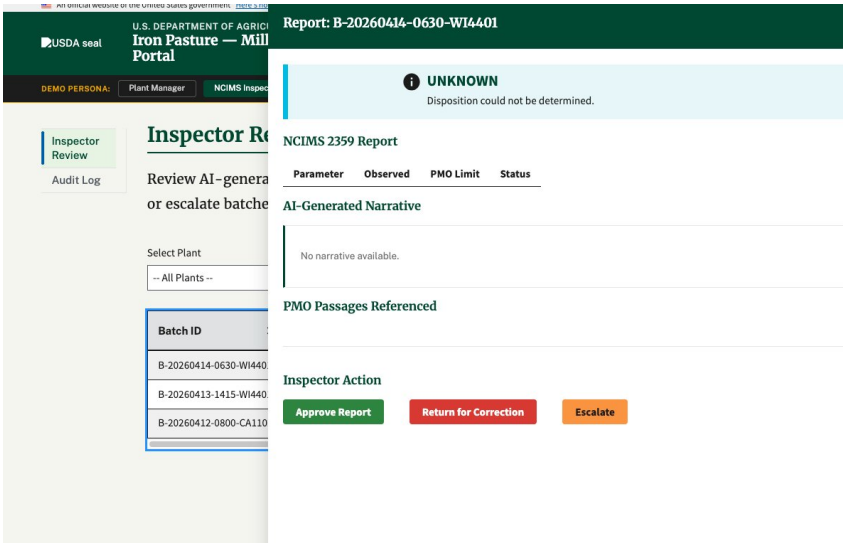
The Plant Manager submits batch sensor readings via the USWDS-compliant form. The Bronze–Silver–Gold pipeline stages are shown with live status indicators. All fields map directly to PMO Section 7 and 16p compliance parameters.

Fig. 2 — Inspector Review Panel (NCIMS Inspector view)



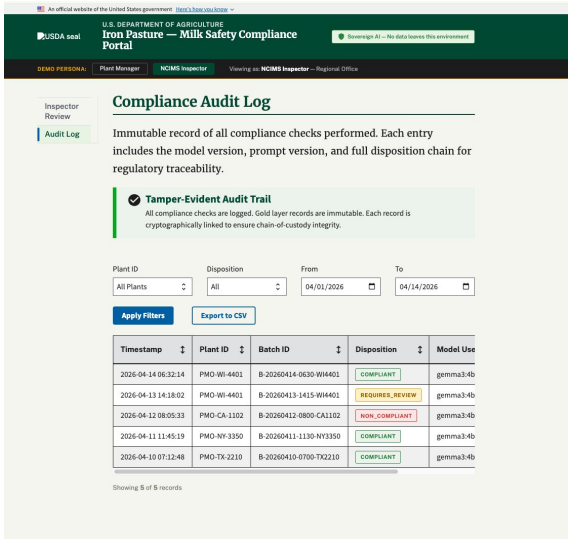
The NCIMS Inspector sees cross-plant compliance history with color-coded disposition badges (COMPLIANT, REQUIRES REVIEW, NON-COMPLIANT). Role separation ensures Plant Managers cannot access this view.

Fig. 3 — NCIMS 2359 Report Detail



Clicking a batch opens the AI-generated NCIMS 2359 report with parameter-level pass/fail status, the LLM compliance narrative, and PMO passages referenced. The inspector can Approve, Return for Correction, or Escalate.

Fig. 4 — Compliance Audit Log



The immutable Gold layer audit log records every compliance check with timestamp, plant ID, disposition, and model version. Date-range and disposition filters support regulatory review. Export to CSV is available for external reporting.